causalgeo: channel effects from concurrent geo holdouts

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Abstract

Advertisers measure media by changing it in some geographic markets and leaving it unchanged in others, then reading the sales gap as the effect of that change. The assignment is chosen by the experimenter, so the contrast does not depend on who happened to see an ad. The design is also expensive. Sales are noisy relative to a typical advertising effect, and a firm can darken only so many markets. A cell large enough to detect one channel can use up that allowance before every channel has been tested, so several channels are turned off together, in several cells, during the same weeks.

That design identifies each bundle. It does not identify the channels inside the bundle. Open libraries for geo experiments estimate one contrast, or several geographic cells, and stop. The channel split is then left to a marketing-mix regression. Those regressions are weak when channels move together, and nothing in the usual workflow checks the split against the experimental total.

`causalgeo` assigns the concurrent cells, reports which channels the off-pattern can name, estimates each bundle from the holdout indicator, and fits channel slopes with that total as a soft calibration. The slopes are pulled toward the bundle effects. They are not forced to equal them, and the bundle table reports the gap that remains. Channels whose impressions are copies of one another are kept as one shared control and are not given their own slopes. On published geo impression paths, with sales built from known slopes, the recovered slopes differ from the planted slopes by at most 0.2 percent.

Keywords: geo experiment, incrementality, holdout, marketing mix, Python

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Table 1: Code metadata.

Nr.	Code metadata description	Value
C1	Current code version	0.1.0
C2	Permanent link to code/repository used for this code version	`https://github.com/adityapt/causalgeo/releases/tag/v0.1.0`
C3	Permanent link to Reproducible Capsule	`https://github.com/adityapt/causalgeo/blob/v0.1.0/paper/reproduce.py`
C4	Legal Code License	MIT
C5	Code versioning system used	git
C6	Software code languages, tools, and services used	Python
C7	Compilation requirements, operating environments and dependencies	Python $\geq$ 3.10. Runtime dependencies: pandas, numpy, scipy, statsmodels. The example also uses matplotlib.
C8	Link to developer documentation	`https://github.com/adityapt/causalgeo/blob/main/README.md`
C9	Support email for questions	puttaparthy.aditya@gmail.com

1. Motivation and significance

Geo experiments assign markets to a treatment or a control and read the sales gap as the effect of the media change [1]. The same design can be rotated across periods when a firm wants a sequence of readings rather than one [2]. Randomized assignment is the point of the method. Observational comparisons of exposed and unexposed users are pulled around by the fact that people who see ads are already doing more of everything else [3], and large platform experiments have found that standard observational models do not recover the experimental answer even with rich covariates [4]. A clean off-switch, as in the eBay paid-search experiment, is still the design that answers whether the channel was buying incremental sales [5].

The practical difficulty is power. Advertising effects are small next to the week-to-week movement in sales, so an informative experiment can demand a very large sample [6]. Analysis has adapted to panels with few units: geo-

based regression uses the cross-section of markets [1], time-based regression fits the pre-test path of the treated markets on the control and reads the test-period gap [7], and Bayesian structural time series builds a synthetic control from the control series [8]. Synthetic control and its extensions do the same job by weighting donor markets to match the treated pre-test path [9, 10, 11, 12]. Synthetic difference-in-differences combines that weighting with unit and time effects [13].

Two open libraries wrap these estimators for advertising. GeoLift selects markets and estimates lift with augmented and generalized synthetic control [14, 12]. Meridian GeoX searches stratified designs and estimates lift with time-based regression, synthetic control, or synthetic difference-in-differences [15, 7, 13]. GeoLift can run more than one treated group. Those groups are geographic cells. They are not coded as channel bundles, and neither library solves a set of bundle totals back into channel slopes. A related software paper makes the same kind of distinction for a different estimand: a unit-level difference-in-differences estimator, rather than one coefficient for every unit treated in the same period [16]. The distinction here is between a bundle the experiment names and a channel it does not.

In a live media program the constraint is often the number of markets that can go dark. A cell large enough to detect one channel can use up the markets that are allowed to go dark before every channel has been tested. The practical design turns a bundle off in each cell, runs the cells in the same weeks, and keeps a shared control. The holdout indicator identifies the bundle total. It does not, by itself, identify the channels inside the bundle.

Marketing-mix models are the usual way to ask for a channel number. They regress sales on channel impressions, with a carryover transform and a saturation curve, and they are fit on observational history [17, 18]. The regression answers a causal question only under strong conditions, and collinear media is one of the conditions it routinely fails: the sum of two channels can be pinned down when the split between them cannot [19]. Impression histories can still split a bundle when the channels do not move together. When they do, the honest output is the bundle, and the channel is left blank. An experiment on the bundle is then a calibration constraint for the mix model, not a substitute for it.

`causalgeo` is a Python package for that job. The experimental object is the cell's holdout indicator. Spend during the test is not the treatment regressor: once the assignment is made, spend is a consequence of the holdout, and putting a post-treatment mediator on the right-hand side changes the

question [20]. The channel slopes are a model. Each measured bundle enters as a weighted calibration row, which pulls the slopes toward that total and leaves a gap the bundle table reports. The package is aimed at analysts who already have a market-level outcome series and channel impression series, and who have to report a channel number from a test that turned channels off in groups.

2. Software description

2.1. Software architecture

The concurrent path is four steps, called through `design_concurrent_test` and `disentangle`.

1. `design.py` draws markets inside strata, builds the cell-by-channel schedule, and reports whether each channel is separate, tied, not tested, or not separable.
2. `identified.py` fits one holdout indicator per cell and withholds a channel coefficient unless a single cell turns that channel off alone.
3. `exposure.py` checks channels that were supposed to stay on. A shift that is large relative to the bundle effect is added as a covariate and the cell is refit.
4. `calibrated.py` fits a linear impression model on the pre-test weeks and on the control markets during the test, and adds one calibration equation per cell.

The cell regression is implemented in `panel_ols.py`. Market means and date means are removed, the grand mean is added back, and the ordinary least squares standard errors are clustered by market [21, 22, 23]. On a balanced panel this is the two-way within estimator [20]. The library's single-cell entry point, `analyze`, is unchanged: a test that was already assigned can be read with time-based regression [7], synthetic control [10], or the same clustered within estimator. That path is supporting code. It is not the contribution. Market selection and power search remain in GeoLift and GeoX [14, 15].

2.2. Software functionalities

Design. The caller passes markets with a stratum label, the channel names, and one list of channels per cell. Within each stratum the markets are shuffled. A share set by `control_share` stays on. The remainder is dealt evenly

across cells. The schedule is an indicator matrix $A_{kc} = 1$ when cell k turns channel c off. The separation report is the rank structure of A. A column of zeros is "not tested." Two identical columns are "tied" with each other. A column whose removal drops the rank is "separate." Any other dependence is "not separable."

Bundle effect. Let $D_{it}^{(k)} = 1$ when market i is assigned to cell k and week t is in the test window. The cell regression is

$$Y_{it} = \sum_k \tau_k \, D_{it}^{(k)} + \alpha_i + \gamma_t + \varepsilon_{it}. \tag{1}$$

The coefficient τ_k is the sales change in cell k during the test, relative to that market's own history and to the common week shocks. A channel receives this coefficient as its own number only when two conditions hold: the separation report says the channel is separate, and exactly one cell's bundle is that channel and nothing else. Overlapping bundles can still span the channel space. They do not justify copying a cell coefficient onto one channel.

Still-on channels. A channel left on in cell k can still differ between the dark markets and the control. The impression shift is the difference in differences

$$S_{kc} = \big(\bar{X}_{c,k,\text{post}} - \bar{X}_{c,k,\text{pre}}\big) - \big(\bar{X}_{c,\text{control,post}} - \bar{X}_{c,\text{control,pre}}\big). \tag{2}$$

A pre-test slope b_c is the two-way-demeaned regression of sales on that channel's impressions before the test. The sales-valued shift is $b_c S_{kc}$. If its absolute value exceeds 10 percent of $|\tau_k|$, the cell is refit against the control only, with those still-on impression series as covariates. The channels that were turned off are not covariates. Their absence is the treatment. The default threshold is the argument `material` and is 0.10.

This control is appropriate when the impression gap was already in the paths. It is the wrong control when the test itself moved budget onto the channels left on: that shift is then part of the total effect of the holdout, and removing it changes the question.

Channel split. Let W_{kc} be the impressions of channel c withheld in cell k,

$$W_{kc} = \max(-S_{kc},\, 0) \tag{3}$$

for channels that cell k turned off; all other entries in that calibration row are zero. Thus the dose is the negative of the same pre/post, test/control impression contrast used by the experiment. The calibration target is $\sum_c \beta_c W_{kc} =$

$-\tau_k$. This is one weighted row in the same regression as the observational block, not an equality constraint. The observational block is the linear model on two-way-demeaned sales and impressions, using every market before the test and the control markets during the test. Dark market-weeks are excluded from that block because they are already used in the calibration rows. Observational rows are weighted by the reciprocal of the residual variance. Each calibration row is weighted by $1/\mathrm{se}(\tau_k)^2$. A precise cell estimate pulls harder than a noisy one. The observational rows can still pull the fitted slopes away from exact reconciliation, so the bundle table reports the remaining gap.

The reported slope covariance is the inverse weighted cross-product of all columns retained in this fit. A copied pair's shared nuisance column remains in that matrix, so its covariance with each reported slope is retained before the nuisance coefficient is hidden. The estimated bundle targets enter through their market-clustered standard errors. The calculation uses a diagonal covariance for those targets; it does not model cross-cell covariance from the shared control or dependence between the target and observational rows. The repeated simulation below assesses this approximation for the example design, rather than treating nominal coverage as guaranteed for every panel.

Two weighted columns are treated as copies when their cosine similarity is above $1 - 10^{-6}$. Dropping both would leave their sales in the residual, where they leak into every channel that remains. One column of the pair stays in the regression as a shared control. Neither member is given a slope. This is the collinearity failure that mix models run into when two media series move together [19]: the calibration row can be satisfied by many splits, and the fit is not allowed to pick one. Remaining columns are kept by a rank check. Daily impact for a kept channel is that week's exposure times $\hat{\beta}_c$. An optional geometric adstock, the standard carryover transform in which yesterday's exposure decays by a constant factor, can be applied to impressions before the fit [17, 18]. The decay defaults to zero, so the default model has no carryover and no saturation curve. Both of those belong in a full mix model [18]. They are not identified by a single on-or-off holdout, which moves each tested channel between its ordinary level and zero rather than tracing a response curve.

The returned tables are the channel slopes, the measured bundle effect next to the effect implied by $\hat{\beta}$ and W_{kc}, the exposure check, and the daily impacts.

3. Illustrative examples

No public geo holdout ships with known channel-level effects. The example uses the published Meridian geo impression file `geo_all_channels.csv`: 40 markets and 156 weeks, from 25 January 2021 through 15 January 2024 [24]. Sales are constructed so the slopes are known. Meridian's conversions column is used only to form strata. It is not the outcome, because that series already contains Meridian's own media response.

Three of the five channels are used, with planted slopes 2×10^{-4}, 5×10^{-4}, and 1×10^{-4} sales per impression. Strata are quartiles of each market's mean conversions. With `control_share = 0.4` and seed 1, the design places 16 markets in control and 8 in each cell. The test window is the last 16 weeks, beginning 2 October 2023. In a dark market-week the turned-off channel's impressions are set to zero, and sales are

$$Y_{it} = \alpha_i + \gamma_t + \sum_c \beta_c X_{itc} + \varepsilon_{it}, \tag{4}$$

with $\alpha_i \sim N(0, 50^2)$, $\gamma_t \sim N(0, 20^2)$, and $\varepsilon_{it} \sim N(0, 30^2)$, drawn with seed 7. The script is `paper/example.py`.

The cells turn off {Channel0, Channel1}, {Channel0, Channel2}, and {Channel1, Channel2}. Written as an indicator matrix, with a one when the channel is off, the three cells are the rows $(1, 1, 0)$, $(1, 0, 1)$, and $(0, 1, 1)$. That matrix has full column rank, so the separation report marks all three channels separate: each column is needed, and removing any one of them loses a direction the cells had covered. Full rank is not the same thing as a pure single-channel test. No cell turns a single channel off alone, so the assignment-only table leaves every channel blank. Copying any one cell coefficient onto one of its two channels would invent a split the holdout did not run. The slopes in Table 3 come from the impression model, using the weeks in which the channels did not move together, softly calibrated using the three bundle estimates.

The planted sales process is linear, which is the model the package fits when adstock is left at zero. The example is therefore a check that the code recovers a slope it is built to recover, on impression paths that were not drawn by the package. It is not a check of saturation, of a long carryover, or of the power of an eight-market cell. Eight markets are a small cell next to the sample sizes advertising experiments often need [6]. The standard errors in Table 2 are wide relative to a decision that would turn on a few percent

Table 2: Cell effects on the constructed series. The adjusted column replaces the cell coefficient when a still-on channel's sales-valued shift exceeds 10 percent of the unadjusted effect. The model column values the same impression change at the fitted slopes: the difference-in-differences in impressions for the channels that cell turned off.

Cell	Bundle	Unadjusted	Adjusted	SE	Model
1	Channel0+Channel1	−555.7	−587.2	71.4	−570.2
2	Channel0+Channel2	−318.0	−251.5	36.2	−253.6
3	Channel1+Channel2	−399.9	−399.9	54.4	−396.2

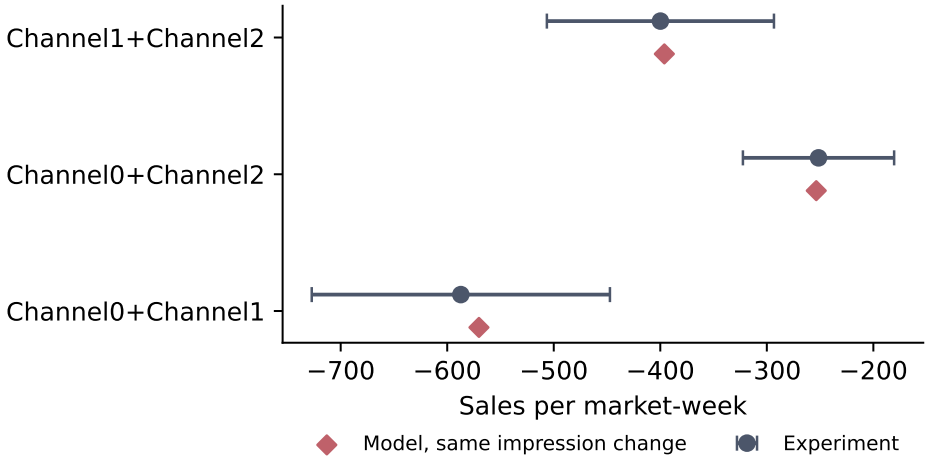


Figure 1: Experimental bundle effect, with a 95 percent interval, and the same off-channel impression change valued at the fitted slopes. Sales per market-week.

of sales. They are narrow relative to the planted effects in this simulation because the planted slopes, applied to these impression levels, move sales by hundreds of units against noise of standard deviation 30. A real holdout with a smaller effect would return the same kind of table and a wider interval.

Cell 2 is the clear exposure adjustment. Channel1 stayed on, and its impression shift was −118,115 against the control, worth −83 sales at the pre-test slope, or 26 percent of the unadjusted cell effect. Controlling for it moves the cell effect from −318.0 to −251.5. Cell 1's still-on Channel2 shift is worth 13 percent of the unadjusted effect, so the rule fires; the coefficient moves from −555.7 to −587.2, a change well inside the standard error of 71. Cell 3's still-on shift is worth 0.5 percent of the cell effect, and the coefficient

Table 3: Planted and recovered slopes, in sales per 10,000 impressions. The largest absolute error is 0.21 percent.

Channel	Planted	Recovered	SE	Abs. error (%)
Channel0	2.000	1.997	0.006	0.17
Channel1	5.000	5.009	0.008	0.17
Channel2	1.000	1.002	0.008	0.21

Table 4: Two hundred refits per setting. Rho is the weight on Channel0 in the untreated Channel1 path. Error is the median absolute percent error among reported slopes. Coverage is the mean within-run coverage; brackets give its 95 percent Monte Carlo interval. At rho 1, error and coverage refer to the remaining channel.

Scale	Rho	Runs	Median error (%)	Coverage [MC interval]	Pair withheld
1.00	0.0	200	0.20	0.958 [0.942, 0.974]	0.00
1.00	0.9	200	0.40	0.953 [0.934, 0.973]	0.00
1.00	1.0	200	0.58	0.940 [0.907, 0.973]	1.00
0.25	0.0	200	0.77	0.958 [0.942, 0.974]	0.00
0.25	0.9	200	1.63	0.957 [0.938, 0.975]	0.00
0.25	1.0	200	2.24	0.940 [0.907, 0.973]	1.00

is left as estimated.

The comparison in Table 2 and Figure 1, written by `paper/figures.py`, uses the exposure contrast in equation (3). The adjusted cell coefficient is the sales gap in equation (1). The model column applies the fitted slopes to the difference-in-differences in impressions for the channels that cell turned off. Those two numbers are −587.2 against −570.2, −251.5 against −253.6, and −399.9 against −396.2. The remaining gaps are 2.9, 0.8, and 0.9 percent of the experimental effects, and each model total sits inside the experimental interval. The bundle coefficient is the experimental result. The channel slope is the model-assisted split.

What the reader should not do is treat a blank channel as a zero. A blank means the design or the impression paths could not name that channel. The sales it may have caused are still inside the bundle total. Reporting a zero would say the channel had no effect. Reporting a blank says the test did not separate it. That is the same reason a mix model should not be forced to print a precise return for two media series that never moved apart [19].

A second run replaces Channel1's impressions with a copy of Channel0

and rebuilds sales from the copied paths. Channel0 and Channel1 are both returned blank. Their shared path stays in the regression as one control, so the sales it generated are not left in the residual. Channel2, whose path is still distinct, is then 0.97 sales per 10,000 impressions against a planted 1.00. Leaving the pair out of the regression entirely pushes that same slope to 3.02. Withholding the split is not enough. The shared path has to be controlled for, or the channels that remain inherit it.

The same Meridian paths were refit at 200 sales-noise seeds per setting, at the planted slopes and at one quarter of those slopes, and with the untreated Channel1 path mixed toward Channel0 before either holdout is applied. This order matters: correlating the paths after applying the holdouts would make one channel's assignment alter another channel that stayed on. Table 4 summarizes `paper/reliability.py`. Figure 2 is drawn from that output by `paper/figures.py`. At the original paths, median absolute error is 0.20 percent and coverage is 95.8 percent. At a mix weight of 0.9, median error is 0.40 percent and coverage is 95.3 percent. Shrinking all slopes by four raises those errors to 0.77 and 1.63 percent, while coverage is 95.8 and 95.7 percent. When Channel1 is an exact copy, both members are withheld in every run. The error plotted at that point is the remaining channel only: 0.58 percent at the original slopes and 2.24 percent at the smaller ones. Its coverage is 94.0 percent; the 95 percent Monte Carlo interval is 90.7 to 97.3 percent. Thus every interval estimate in this grid is statistically compatible with 95 percent coverage. This is evidence for this design and data-generating process, not a general proof of the diagonal covariance approximation.

4. Impact

The package gives a team that must test channels in bundles one place to assign the cells, see which channels the design can name, read the bundle effects from the holdout indicators, and produce channel slopes with the remaining gap reported next to those effects. Existing open geo libraries stop at the cell [14, 15]. Analysts who need a channel number have had to do the split in a separate mix model [19, 18], with no report of how far the split sits from the experimental total, and no refusal when two channels moved together. The same separation of a fine estimand from a coarse one is what recent panel software has started to make explicit [16]. Here the coarse object is the bundle, and the fine object is the channel.

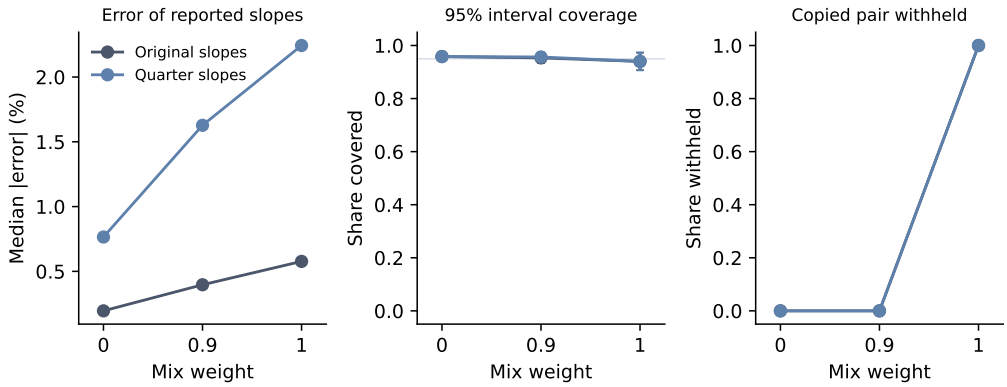


Figure 2: Recovery over 200 simulations per setting as the untreated Channel1 path is mixed toward Channel0. Bars in the coverage panel are 95 percent Monte Carlo intervals. At mix weight 1 the pair is withheld in every run, so error and coverage refer to the remaining channel.

The current release is 0.1.0. Adoption is early, and this paper does not report a user study. The reusable piece is the example: `paper/reproduce.py` rebuilds Tables 2, 3, and 4, and both figures. The same functions apply to a real holdout once the caller supplies the market outcome and the channel impressions. Channels the design cannot separate, and channels whose impressions are copies, stay blank, which is the output a downstream budget decision can tell from a slope that was actually estimated.

5. Conclusions

`causalgeo` measures concurrent channel-bundle holdouts and splits each measured total into channel slopes when the impression histories support the split. The calibration is soft and uses the experiment's pre/post impression contrast. On the published Meridian paths the three slopes are recovered to within 0.2 percent, and their bundle totals are 0.8 to 2.9 percent from the experimental effects. A copied pair is controlled for as one path and is not split. Across 200 simulations per setting, the channel that remains is close to its planted slope and its nominal interval covers in 94 percent of runs. Coverage across the six settings ranges from 94.0 to 95.8 percent, with Monte Carlo intervals that include 95 percent. These intervals remain conditional on the fitted linear model and the diagonal approximation for estimated bundle targets.

The split is linear, with an optional geometric adstock and no saturation curve [18]. A holdout that switches a channel off does not trace the curve between those two points, so the package does not estimate one. Market-selection and power search are left to GeoLift and GeoX [14, 15]; this package assigns markets inside strata the caller already chose, and it does not claim that eight markets are a powered cell [6]. The still-on adjustment should be read against the design of the test: it removes a pre-existing impression imbalance, and it should be turned off when the holdout itself is what moved the budget. Spend during the test stays out of the regression, because it is a result of the assignment rather than a confounder of it [20, 3].

CRediT authorship contribution statement

Aditya Puttaparthi Tirumala: Conceptualization, Methodology, Software, Validation, Writing – original draft, Writing – review and editing.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The impression paths are the Meridian file `geo_all_channels.csv` [24]. From `paper/` in `https://github.com/adityapt/causalgeo`, `PYTHONPATH=../src python3.11 reproduce.py` rebuilds Tables 2, 3, and 4 and both figures, then checks them against the rounded values reported here. Table 1 is manuscript metadata.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Cursor, with the Grok language model, to draft and revise the manuscript and the example code. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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